GLD/GDX Pairs Trading

Cointegration Analysis & Kalman Filter Spread

Period: 2010-2024 | Source: Yahoo Finance via yfinance

1. Dataset & Motivation

GLD (SPDR Gold Trust) and GDX (VanEck Gold Miners ETF) are two of the most liquid gold-linked instruments available. GLD tracks spot gold directly; GDX holds a basket of gold mining companies, which means it is driven by gold prices but also carries equity-specific risk -- management decisions, operating costs, capital structure, and sector sentiment all affect GDX independently of where gold trades.

Because both instruments share the same underlying commodity, their prices should move together over long horizons. If that relationship is stable and mean-reverting, deviations from it can be traded: buy the underperformer, short the outperformer, and wait for the gap to close. The goal of this project is to test whether that relationship holds over 2010-2024, understand when and why it breaks down, and design a signal that only fires when the statistical evidence supports it.

2. Methods Overview

The analysis runs through five main stages:

- **Data collection & EDA**
  Daily adjusted close prices from Yahoo Finance. Normalized price charts, rolling correlation, volatility, and return distributions built up intuition before any formal testing.
- **Stationarity testing (ADF)**
  Augmented Dickey-Fuller tests confirmed both log price series are I(1) -- non-stationary in levels, stationary in first differences. Required before cointegration testing.
- **Cointegration testing**
  Engle-Granger and Johansen tests on the full period, then repeated on rolling 2-year windows to map when the relationship was active vs. broken down.
- **Kalman filter**
  Rather than a fixed hedge ratio, the Kalman filter updates its estimate at every timestep as new prices arrive -- continuously re-learning the relationship rather than assuming it's fixed.
- **Signal generation**
  Z-score thresholds on the Kalman spread, gated by the rolling cointegration test so signals only fire when the statistical relationship is confirmed active.

3. Key Findings

The headline result is that GLD and GDX are not cointegrated over the full 2010-2024 period (Engle-Granger p = 0.45, Johansen trace = 10.59 vs. 15.49 critical). This is not a methodological failure -- it is the most interesting finding. GLD approximately doubled over 14 years while GDX declined roughly 40% from its 2011 peak. That permanent divergence cannot be closed by a mean-reverting spread.

- **Regime-dependent relationship**
  The pair was cointegrated in roughly 79% of rolling 2-year windows despite failing the full-period test. The relationship was real and tradeable for extended stretches, but macro forces -- most severely the 2013 gold crash and the 2022 rate hike cycle -- periodically broke it down.
- **Static hedge ratio is unstable**
  Rolling OLS estimates of the hedge ratio ranged from -0.10 to 1.15 across the sample. A single static estimate of 0.34 sits in the middle of that range but is wrong for most individual periods. This directly motivates a time-varying approach.
- **Kalman filter resolves the non-stationarity**
  The dynamic spread achieved an ADF p-value of <0.0001 vs. 0.87 for the static spread -- a clear improvement. The filter's hedge ratio tracked the rolling OLS beta closely but without the 504-day lag of a fixed rolling window.
- **ARIMA out-of-sample accuracy**
  An ARIMA(1,0,1) model on the Kalman spread outperformed a naive random walk by 23.3% on the holdout period (RMSE 0.00144 vs. 0.00188), confirming short-term autocorrelation structure that z-score signals can exploit.
- **Regime-aware signals are more coherent**
  110 trades were generated over the full period (59 long, 51 short), with 88.7% of days flat or gated. Signals were suppressed during non-cointegrated windows -- periods where a naive z-score approach would have traded into a spread with no mean to revert to.

4. Key Figures

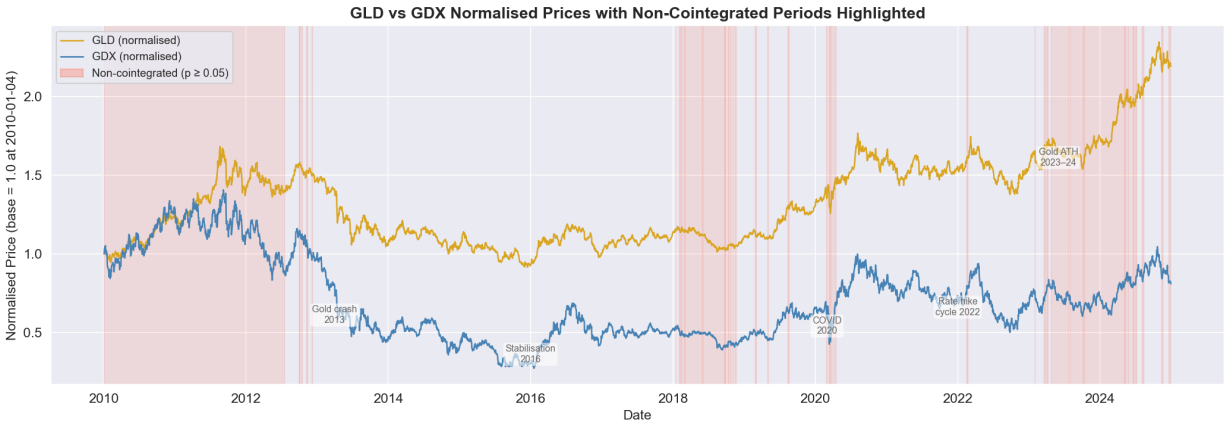

Figure 1: GLD and GDX normalised prices (2010-2024). Red shading marks non-cointegrated windows (rolling EG p >= 0.05). The persistent divergence post-2013 drives the full-period result.

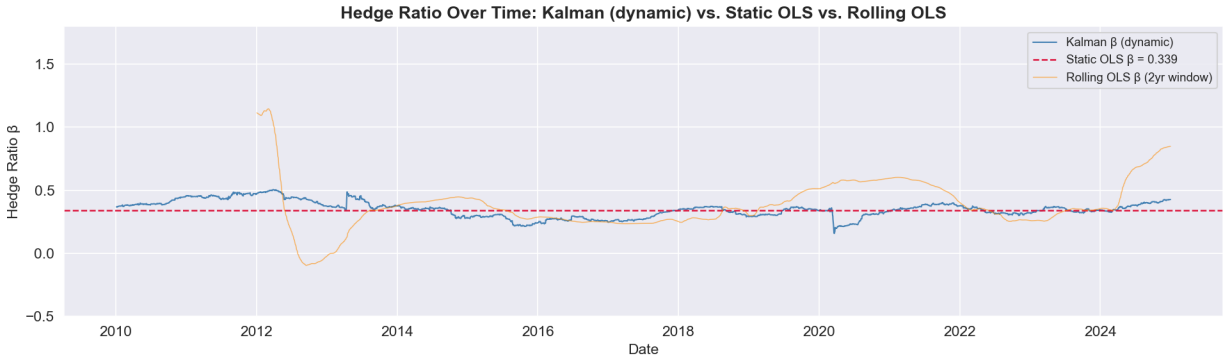

Figure 2: Hedge ratio over time. Flat red = static OLS (0.34). Blue = Kalman dynamic (0.16-0.51). Orange = rolling OLS beta (2-year window).

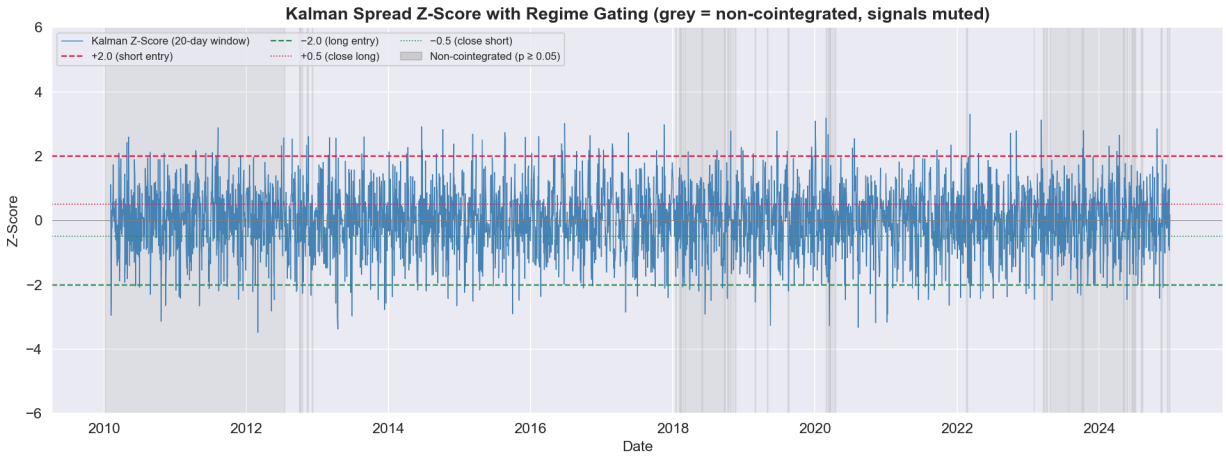

Figure 3: Kalman spread z-score with regime gating. Grey shading = non-cointegrated periods where signals are suppressed. Entry at +/-2 sigma, close at +/-0.5 sigma.

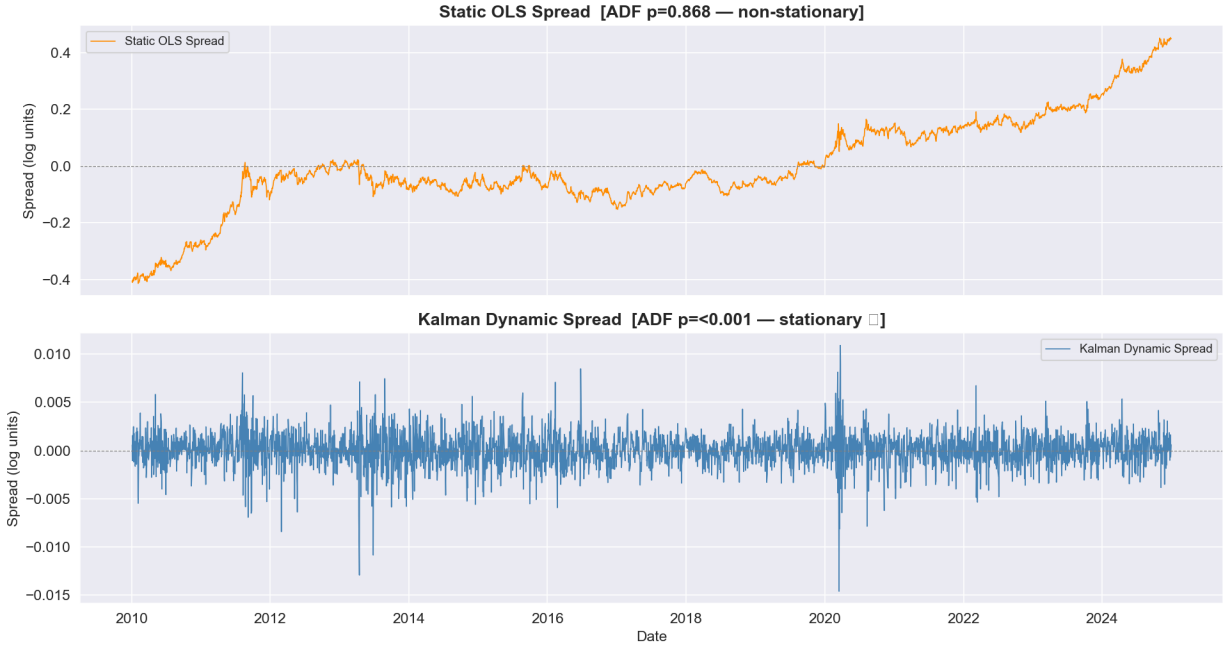

Figure 4: Static OLS spread (top, ADF p = 0.87) vs. Kalman dynamic spread (bottom, ADF p < 0.0001).

5. Limitations & Reflection

A few important caveats on what this analysis does and doesn't show:

- **Look-ahead bias in initialisation**
  The Kalman filter is warm-started with the full-period OLS beta. In a live setting you wouldn't have that estimate upfront -- a burn-in period would be needed, introducing mild look-ahead bias in the earliest part of the sample.
- **No transaction costs**
  The analysis counts entries and exits but doesn't model bid-ask spread, market impact, or borrowing costs for the short leg. Short ETF costs are non-trivial for multi-month holds.
- **Thresholds not optimised**
  The +/-2 sigma entry and +/-0.5 sigma close thresholds are standard but not calibrated to this pair. In-sample optimisation risks overfitting given the limited number of independent regime windows in the sample.
- **Recent non-cointegration**
  Rolling windows in 2022-2024 show weak or absent cointegration. Under the regime gate, the strategy would currently be flat -- the statistical foundation for a live trade is not present in the most recent data.
- **No PnL or Sharpe ratio**
  This analysis evaluates whether the statistical foundation for a pairs trade exists, not whether it's profitable. Return attribution, position sizing, and risk-adjusted performance are outside scope.